

Decision and Risk Analysis

Lecture 4: **Directed Acyclic Graphs, Bayesian Balance and Sequential Learning**

DAEN 427 / ISEN 427 / ISEN 627

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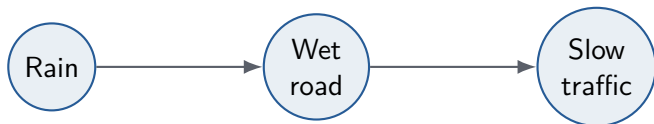
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The basic idea

A **directed acyclic graph**, usually called a **DAG**, contains:

- ▶ **Nodes**: things we care about;
- ▶ **Arrows**: directed relationships;
- ▶ **No directed cycles**: we cannot follow arrows and return to where we started.

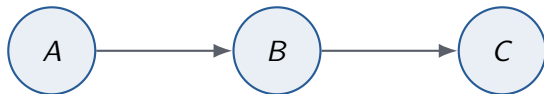


Informally, we read this as

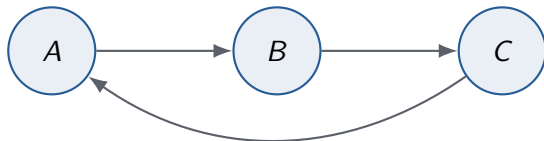
Rain $\longrightarrow$ Wet road $\longrightarrow$ Slow traffic.

Why “acyclic”?

This is a DAG:



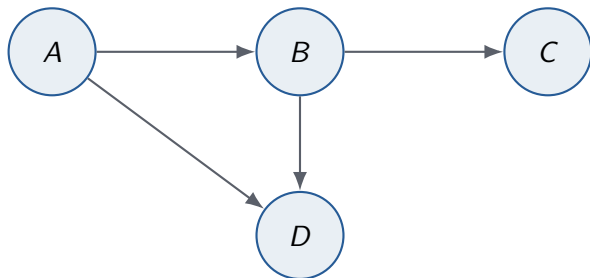
But this is **not** a DAG:



Following the arrows produces the directed cycle

$$A \longrightarrow B \longrightarrow C \longrightarrow A.$$

Some family vocabulary



Relative to B :

- ▶ A is a **parent** of B ;
- ▶ C and D are **children** of B ;
- ▶ A is an **ancestor** of C and D ;
- ▶ C and D are **descendants** of A .

A valid **topological ordering** is A, B, C, D .

What does an arrow mean?

An arrow $X \rightarrow Y$ does not have one universal meaning.

Depending on the application, it may represent

- ▶ a computational dependency;
- ▶ a prerequisite;
- ▶ a temporal relationship;
- ▶ direct probabilistic dependence;
- ▶ or a causal effect.

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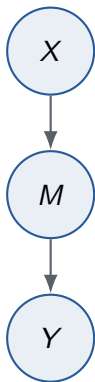
Important warning

A DAG is causal only when the arrows are assigned a causal interpretation and the required assumptions are defensible.

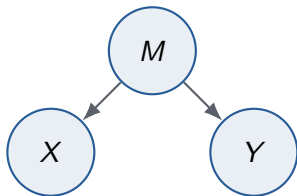
Correlation alone does not tell us which way an arrow should point.

Three patterns worth remembering

Chain

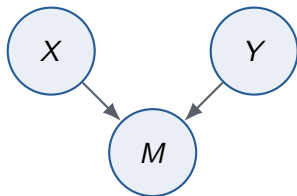


Fork



$$X \leftarrow M \rightarrow Y$$

Collider



$$X \rightarrow M \leftarrow Y$$

$$X \rightarrow M \rightarrow Y$$

These patterns behave differently when we condition on M .

Conditioning can block or open a path

Pattern	Normally	Given M
$X \rightarrow M \rightarrow Y$	Path open	Path blocked
$X \leftarrow M \rightarrow Y$	Path open	Path blocked
$X \rightarrow M \leftarrow Y$	Path blocked	Path opened

The collider is the surprising case.

Informal example

Ability and family connections may both affect admission:

$$\text{Ability} \longrightarrow \text{Admission} \longleftarrow \text{Connections.}$$

Among admitted students, ability and connections may appear related even if they are unrelated in the general population.

D-separation, informally

D-separation is a graphical test for deciding whether two sets of variables are conditionally independent.

A path is blocked when:

1. it contains a chain or fork whose middle node is conditioned on; or
2. it contains a collider, unless the collider or one of its descendants is conditioned on.

If every path between X and Y is blocked by Z , then the DAG implies

$$X \perp\!\!\!\perp Y \mid Z.$$

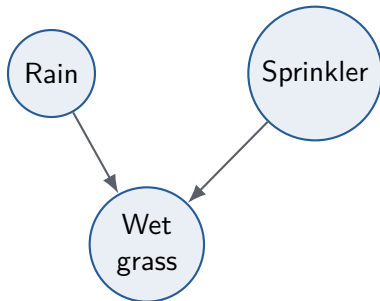
Plain-language interpretation

After we know Z , learning X gives us no additional information about Y .

From a DAG to a Bayesian network

A **Bayesian network** combines:

1. a DAG describing local dependencies; and
2. a probability distribution for each node given its parents.



The joint distribution factorizes as

$$P(R, S, W) = P(R) P(S) P(W \mid R, S).$$

The general factorization

For variables $X_1, \dots, X_n$, a Bayesian network represents

$$P(X_1, \dots, X_n) = \prod_{i=1}^n P(X_i \mid \text{Pa}(X_i)),$$

where $\text{Pa}(X_i)$ denotes the parents of X_i .

Why is this useful?

Instead of specifying one enormous joint probability distribution, we specify several smaller conditional models.

For binary variables, an unrestricted joint distribution over n variables can require as many as

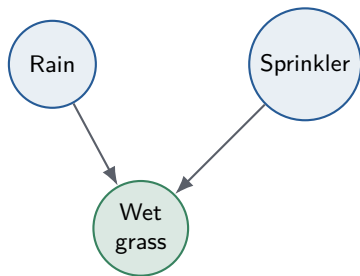
$$2^n - 1$$

independent parameters.

A sparse Bayesian network can require dramatically fewer.

Evidence flows through the network

Suppose we observe that the grass is wet.



Bayes' rule updates our beliefs:

$$P(R, S \mid W) \propto P(W \mid R, S)P(R)P(S).$$

- ▶ Wet grass raises the probability of rain.
- ▶ It also raises the probability that the sprinkler ran.
- ▶ Learning that it rained may reduce the need to explain the wet grass using the sprinkler.

The last effect is called **explaining away**.

Prediction, diagnosis, and intervention

The same graph can support different kinds of questions.

Prediction: If it rains, how likely is wet grass?

$$P(W \mid R)$$

Diagnosis: If the grass is wet, how likely was rain?

$$P(R \mid W)$$

Intervention: What happens if we deliberately turn on the sprinkler?

$$P(W \mid \text{do}(S = 1))$$

The first two are probabilistic queries.

The third is causal: it changes the data-generating process instead of merely observing it.

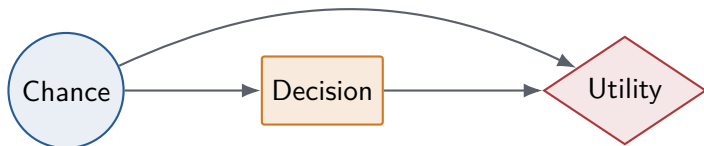
A decision needs more than probabilities

A Bayesian network tells us what is uncertain and how beliefs change.

A decision problem also requires:

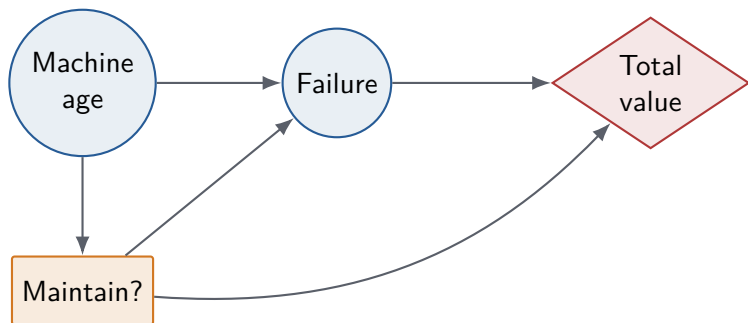
- ▶ the available **actions**;
- ▶ the **information** available when choosing;
- ▶ and a definition of what outcomes are **valuable**.

An **influence diagram** extends a Bayesian network.



circles = uncertainty, rectangles = choices, diamonds = value.

Example: should we perform maintenance?



- ▶ Machine age is known before the decision.
- ▶ Maintenance changes the probability of failure.
- ▶ Maintenance costs money.
- ▶ Failure may cost much more.

The goal is not simply to predict failure. The goal is to choose the action with the best expected consequences.

Expected utility: the decision rule

Let d be a decision and let x be an uncertain outcome.

$$EU(d) = \sum_x P(x \mid d, e) U(d, x),$$

where e denotes the available evidence.

Choose

$$d^* = \arg \max_d EU(d).$$

In plain language

For each decision:

1. consider the possible outcomes;
2. weight each outcome by its probability;
3. attach a value to each outcome;
4. choose the decision with the largest expected value.

A small maintenance calculation

Suppose preventive maintenance costs \$2,000.

Without maintenance:

$$P(\text{failure}) = 0.20$$

and a failure costs \$20,000.

The expected cost without maintenance is

$$0.20(\$20,000) = \$4,000.$$

If maintenance essentially prevents failure, then:

$$E[\text{cost} \mid \text{no maintenance}] = \$4,000,$$

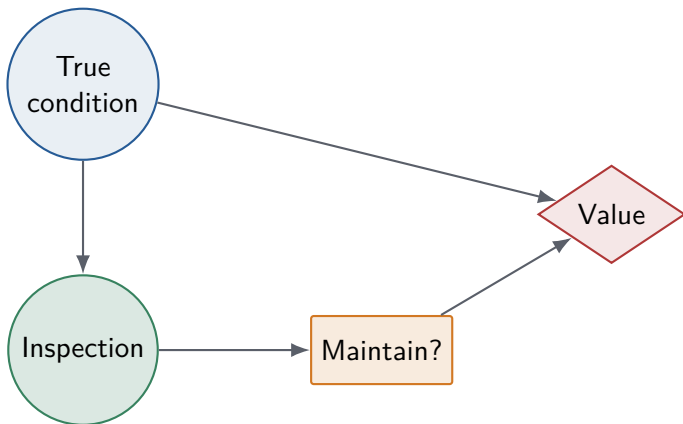
$$E[\text{cost} \mid \text{maintenance}] = \$2,000.$$

Decision

On these assumptions, preventive maintenance is preferred.

Information itself can have value

Suppose an inspection provides information before maintenance is chosen.



The **value of information** is

$$VOI = EU(\text{best policy with information}) - EU(\text{best policy without information})$$

A practical modeling workflow

1. Define the decision and the objective.
2. List the relevant uncertainties and observations.
3. Draw arrows representing direct relationships.
4. Check that the graph is acyclic.
5. Specify conditional probability models.
6. Specify utilities, benefits, or costs.
7. Enter evidence and update probabilities.
8. Compare actions using expected utility.
9. Test sensitivity to uncertain assumptions.

A useful modeling habit

Start small. Add a node only when it could change beliefs, decisions, or the interpretation of the model.

Common pitfalls

- ▶ Treating every arrow as automatically causal
- ▶ Omitting an important common cause
- ▶ Conditioning on a collider without realizing it
- ▶ Adding so many variables that the model becomes unmanageable
- ▶ Using precise probabilities that are not well supported
- ▶ Optimizing predictions instead of the actual decision objective
- ▶ Forgetting when information becomes available
- ▶ Reporting one “optimal” answer without sensitivity analysis

A useful principle

A good decision model should be understandable enough for its assumptions to be questioned.

Takeaways

1. A DAG is a directed graph with no directed cycles.
2. Its structure encodes relationships and possible independencies.
3. A Bayesian network adds local probability models.
4. Conditioning can block paths or open collider paths.
5. An influence diagram adds decisions and utilities.
6. Decision analysis chooses actions using expected utility, not probability alone.

DAGs turn a complicated story into a model that we can inspect, calculate with, and improve.